MOLMO-SCENEGRAPH: ADDING STRUCTURE TO PRETRAINING YIELDS STRONG VISION-LANGUAGE MODELS

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ABSTRACT

Pretraining on large, diverse collections of image-caption pairs is the standard method for training strong Vision-Language Models (VLMs). However, captions have several drawbacks due to their lack of consistent structure: they can be brief, ambiguous, and leave out information that annotators think is "obvious", leading to failures on downstream grounding and reasoning tasks. Motivated by human visual processing, we instead use a *structured* scene description: scene graphs. We curate PixMo-SceneGraph, a dataset of 704K image-scene-graph pairs derived solely from captions in the PixMo-Cap dataset, enabling a controlled comparison with unstructured caption data. We pretrain Molmo2 on PixMo-SceneGraph and finetune the resulting models on 8 standard academic datasets, finding that scene-graph pretraining improves performance over caption pretraining, particularly in generalizing to the grounding and reasoning tasks where caption-based models struggle. Combining scene graphs with captions yields further gains. Moreover, the structure of scene graphs also provides an interface for targeted data augmentation: we augment PixMo-SceneGraph with information typically omitted from captions—temporal statements, negations, affordances, and external knowledge—further improving performance. Our suite of models, Molmo-SceneGraph, outperform Qwen3-VL at 2B, 4B, and 8B scales across a broad range of academic, grounding, and reasoning benchmarks, despite being trained on 200–500$\times$ fewer tokens. Our results highlight the value of structure in pretraining data, and encourage researchers to rethink standard methods for VLM pretraining.

1 INTRODUCTION

Modern vision-language models (VLMs) (Radford et al., 2021; Liu et al., 2023b; Bai et al., 2023) are pretrained on large and diverse collections of image-caption pairs (Schuhmann et al., 2022). However, captions have fundamental limitations as a source of supervision: they are often brief, ambiguous, and systematically omit information that people consider "obvious", e.g., spatial relations and notions of causality (Gordon & Van Durme, 2013; Misra et al., 2016). As a result, important aspects of visual data are under-represented in pretraining corpora, limiting the ability of VLMs trained on them to perform important tasks such as grounding and reasoning (Kamath et al., 2026; Al-Tahan et al., 2024). This raises a question: *can we design supervision to encourage representations that (1) are dense and unambiguous; and (2) incorporate information omitted by captions?*

We posit that introducing *structure* into VLM pretraining data leads to better representations. We pretrain VLMs on a structured image representation: scene graphs. Scene graphs instantiate our desire for *unambiguity* by providing explicit and consistent dense supervision for grounded objects, attributes, and relations (Krishna et al., 2017; Park et al., 2025). Further, they more closely align with how humans think and perceive the world (Behrens et al., 2018; Hafri & Firestone, 2021).

We curate PixMo-SceneGraph, a dataset of 704K image-scene-graph pairs, by synthetically converting 704x captions from PixMo-Cap (Deitke et al., 2024) into scene graphs. To isolate the effect of structure, i.e., *unambiguous* information organization, the scene graphs are synthesized without access to the images. We serialize each graph as a JSON with explicit object identities, attributes, relations, and counts, and use these targets to pretrain the Molmo2 VLM (Clark et al., 2026). After

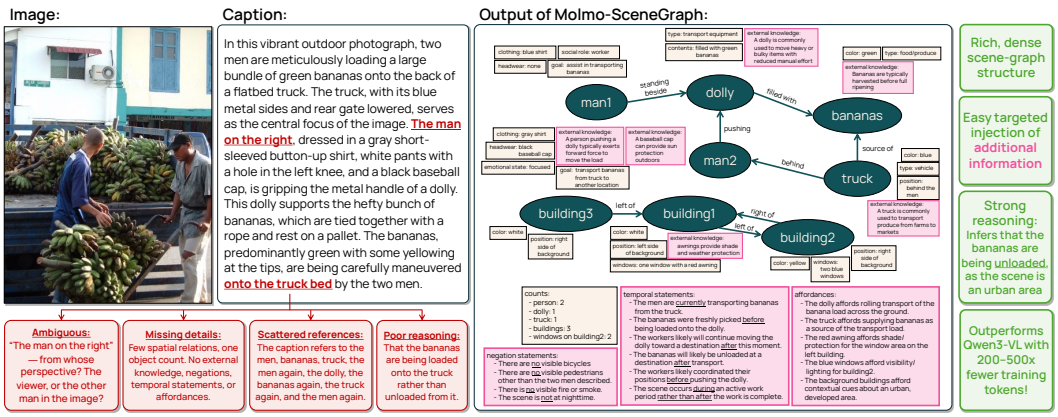


Figure 1: Molmo-SceneGraph is a powerful VLM pretrained on image-scene-graph pairs rather than image-caption pairs. The structured data allows for dense supervision and targeted data augmentation with information typically left out of captions, resulting in a model with strong reasoning capabilities that outperforms Qwen3-VL despite being trained on only 704K images.

pretraining, caption- and scene-graph-pretrained variants are post-trained with identical downstream data from 8 academic benchmarks. We call this suite of models Molmo-SceneGraph. Figure 1 shows an abbreviated scene graph output of Molmo-SceneGraph in comparison to a PixMo-style caption for the depicted image. The generated scene graph is dense, and while some relations in the caption are ambiguous ("the man on the right"—from whose perspective?), the scene graph is explicit ("building1 $\xrightarrow{\text{left of}}$ building2").

We then study the second part of our earlier question: once this structure is available, *can it serve as a structured interface for adding information that captions systematically omit?* We exploit the scene graph schema as an interface for targeted augmentation and add new fields to the pretraining data that enables Molmo-SceneGraph to generate, as seen in Figure 1, external knowledge ("Bananas are typically harvested before full ripening"), temporal inferences ("The bananas will be unloaded at a destination after transport"), negations ("There are no visible bicycles in the image"), and affordances ("The dolly affords rolling transport of the banana load across the ground")—topics that VLMs typically struggle with due to omission from pretraining corpora (Kamath et al., 2026). By systematically adding or removing these types of supervision, we can test not only whether structure helps, but which additional forms of information improve grounding and reasoning beyond the core graph: e.g., while the caption in Figure 1 incorrectly assumes that the men are loading bananas *onto* the truck, Molmo-SG's scene graph correctly reasons that the men are unloading bananas *from* the truck, as trucks are commonly used to transport produce from farms to markets (a generated external knowledge fact), and the scene takes place in an urban area (a generated affordance).

We evaluate our model on the 8 academic benchmarks it was finetuned on, 3 general-purpose benchmarks, and 4 benchmarks that test grounding and reasoning: spatial reasoning with VSR (Liu et al., 2023a), temporal inference with VisualCOMET (Park et al., 2020), Reporting Bias probes (Kamath et al., 2026), and object hallucination with POPE (Li et al., 2023). Relative to caption-only pretraining, scene-graph-only pretraining approximately preserves the standard average while slightly improving the out-of-domain average at the 2B and 4B scales. Combining graphs with captions improves these averages further, and targeted augmentation further benefits out-of-domain grounding and reasoning (Tables 1 and 2). This model exceeds the in-domain average of Qwen3-VL (Bai et al., 2025)—which uses the same text and vision backbones, but is trained on over 200x more tokens—at 2B, 4B, and 8B scales by 4.4 points, 2.9 points, and 1.7 points respectively, and its out-of-domain average at 2B and 4B by 2.0 points and 0.8 points; Qwen3-VL remains stronger out of domain at 8B by 0.6 points. Molmo-SceneGraph also outperforms LLaVA-OneVision-1.5 8B (An et al., 2025) and LLaVA-OneVision-2.0 8B (An et al., 2026), as well as Qwen3.5 9B (Team, 2026) in-domain. Further, our model outperforms the Molmo2 (Clark et al., 2026) finetuned model in-domain and out-of-domain, highlighting the value of our structured pretraining data.

Taken together, our results show that VLM pretraining benefits not only from *more* supervision, but from supervision that makes visual *structure* explicit. Structured targets preserve relationships that captions often leave implicit, provide a natural interface for adding new supervision, and improve grounding and reasoning without requiring more image-text data—outperforming models trained on substantially more data. These results suggest that the next step in VLM pretraining may be to rethink not just the scale of image-text corpora, but the form of the supervision itself. We will release all of our datasets and models in the spirit of open research.

2 Related Work

Vision-Language Model Training. Modern vision-language models (VLMs) commonly combine a pretrained vision encoder with a pretrained LLM through a learned connector, followed by multimodal instruction tuning (Liu et al., 2023b; Bai et al., 2023). The connector is typically trained to align visual representations with the LLM's language space using large-scale image-text data, often drawn from web-scale corpora such as LAION-5B (Schuhmann et al., 2022). After this, the model is finetuned on task-specific data, such as academic benchmarks and/or user inputs: e.g., LLaVA uses instruction tuning (Liu et al., 2023b); Qwen-VL uses a visual receptor and a multi-stage training pipeline with large-scale multimodal data (Bai et al., 2023); and Molmo pretrains on PixMo-Cap (Deitke et al., 2024) before finetuning on a combination of academic datasets, synthetic data, and pointing data. We use the Molmo2 architecture (Clark et al., 2026) to train Molmo-SceneGraph.

Structured Supervision for VLMs. Several works have explored incorporating structured supervision into vision-language learning to improve compositional and relational understanding. Yu et al. (2021) introduces object, attribute, and relationship prediction objectives derived from scene graphs, while Herzig et al. (2023) incorporate scene graphs into pretrained VLMs through an additional scene-graph component and fine-grained captions. More recently, Synthetic Visual Genome (Park et al., 2025) builds on Visual Genome (Krishna et al., 2017) by using language models to densely augment scene-graph annotations and train a multimodal model on the resulting data, demonstrating the value of dense relational supervision for visual reasoning. In contrast, we investigate scene graphs as the primary textual representation for VLM pretraining, rather than as auxiliary supervision or an augmentation of existing image-caption training.

Information Content and Reporting Bias in VLM Data. Beyond the form of supervision, recent work has examined how the information content of VLM training data affects downstream capabilities. Doveh et al. (2023) show that both caption quality and density are important for compositional reasoning. Similarly, the PixMo-Cap dataset (Deitke et al., 2024) collects detailed human descriptions and demonstrates the effectiveness of high-quality, dense image-text data for VLM training. However, even detailed captions can systematically omit information that people consider obvious. Kamath et al. (2026) study this phenomenon as reporting bias (Gordon & Van Durme, 2013; Misra et al., 2016) and show that such omissions can limit downstream reasoning capabilities. Our work builds on these observations by asking whether structuring caption information as scene graphs and explicitly incorporating otherwise omitted information can improve VLM performance.

3 Adding Structure to VLM Pretraining

3.1 Why Structure, and Why Scene Graphs?

Natural language captions have two primary drawbacks in representing a visual scene. First, natural language is ambiguous: e.g., "left of the woman" in an image can refer to completely different positions depending on whether the woman's perspective is considered, or the viewer of the image (Zhang et al., 2025). Second, when writing captions, people (and models trained on human annotations) exhibit a reporting bias and systematically omit "obvious" or "irrelevant" information: e.g., spatial and temporal relations, negations, and counts (Kamath et al., 2026). Both of these drawbacks can be attributed to the unstructured nature of image captions: as there is no consistent slot for each type of information, what information to include is left to convention and inference (Von Stutterheim & Klein, 1989) rather than made explicit. The promise of structured representations is that a consistent, labeled format can both resolve ambiguity and guide information inclusion.

Structured representations separate content from structure, rather than compressing both into a single linear description. Following earlier work in this space (Krishna et al., 2017), we draw inspiration from how the brain processes images. Human visual representations are not just descriptions of an image; they preserve structured information that other cognitive systems such as memory, language, and motor control can access (Cavanagh, 2011), and they separate the processing of object identity from that of spatial location and action (Mishkin et al., 1983; Goodale & Milner, 1992). Perception also captures relations between objects, either automatically as part of seeing, or computed later through deliberate reasoning (Hafri & Firestone, 2021). Visual representations keep different kinds of information distinct rather than collapsing into a single description, similar to how a scene graph separates nodes from other nodes and edges.

The brain's memory system offers further reason to select scene graphs as a structured representation. Evidence suggests that the brain uses a relational encoding format, representing relationships among elements, not just the elements themselves (Biederman et al., 1982; Eichenbaum & Cohen, 2014; Constantinescu et al., 2016; Behrens et al., 2018). Computational modeling proposes that this format enables generalization to new environments because the relational graph is learned separately from sensory content (Whittington et al., 2020). In other words, factorizing structure from content enables transfer to new environments with entirely new observations.

These findings from neuroscience and cognitive psychology motivate representations that keep the content of a scene distinct from the relationships among its elements. We apply this principle to vision-language learning by representing scenes as graphs that explicitly encode objects and their relations: e.g., "cup$\xrightarrow{\text{on}}$table" and "book$\xrightarrow{\text{on}}$shelf" share relational structure despite differing in content. Unlike captions, where relations must be inferred from linguistic context, scene graphs make them explicit, giving the model direct supervision on what is in a scene and how its elements relate.

3.2 Curating PixMo-SceneGraph

Having selected scene graphs as embodying the type of structure desired in our pretraining data, we now curate PixMo-SceneGraph: a corpus of 704K synthetically-generated scene graphs for VLM pretraining. We source our images and captions from PixMo-Cap (Deitke et al., 2024), which contains 712K diverse, high-quality image-caption pairs annotated by humans.

Generating scene graphs. To enable a controlled comparison between caption-based and scene-graph-based pretraining, we generate each scene graph using only the corresponding PixMo-Cap caption, *not the image itself*. This ensures that the scene graph does not contain information about the image unavailable in the caption, and prevents the scene-graph generation process from effectively distilling information from a proprietary vision-language model. We provide GPT-5.4-nano with the original caption and a detailed prompt instructing it to convert the caption into a structured scene graph in the form of a JSON dictionary. The prompt specifies the schema of the scene graph and provides strict instructions to preserve all information contained in the original caption, and is provided in the Appendix. We discard PixMo-Cap examples for which the original image is no longer retrievable from its URL.

Each scene graph consists of four primary components: a dictionary of objects with an entry for each object alongside a unique object ID, object name, and associated attributes; a dictionary of relationships between objects, referring to objects by their unique IDs; a dictionary of object counts by mapping object types to their corresponding counts; and an additional notes field capturing information from the caption that does not naturally fit into the other fields. A scene graph abbreviated for space is shown in Figure 2, with the full scene graph provided in the Appendix.

Augmenting scene graphs. The structured representation also allows us to systematically augment the scene graphs with additional information. To avoid requiring separate data-generation passes, we prompt GPT-5.4-nano to generate several types of auxiliary information alongside the core scene graph, while restricting each type to a specifically designated dictionary key, which are used during postprocessing to construct separate versions of the training data with and without the additional information. We generate external knowledge about objects (two facts per object), temporal statements about the image, negation statements, and affordances. We additionally provide the option to generate information about agents in the caption and their potential goals, as well as external knowledge about relationships between objects (c.f. Fig. 2). Postprocessing details c.f. Appendix.

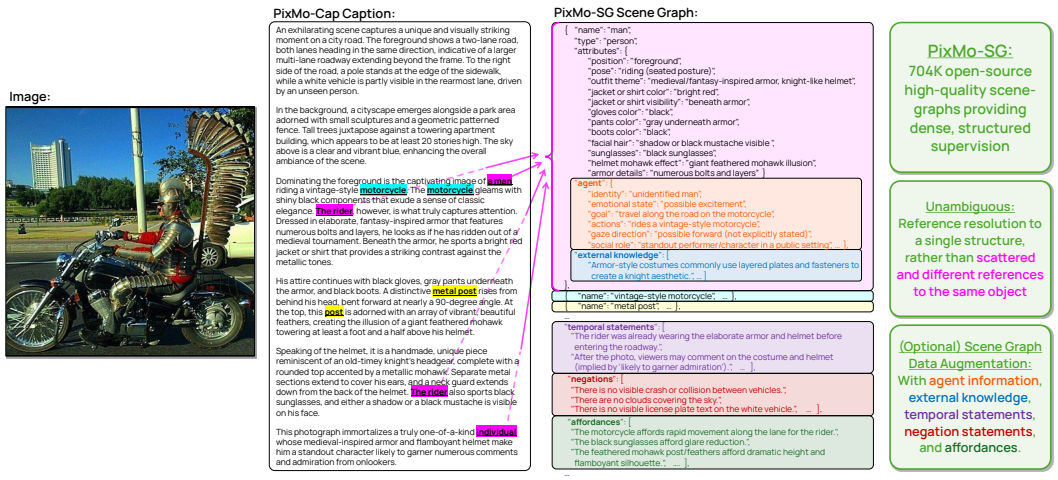


Figure 2: An example from the PixMo-SceneGraph pretraining corpus (abbreviated for space), alongside the PixMo-Cap caption. The scene graph structure allows for unambiguous reference resolution, as well as optional data augmentation of information typically omitted from captions.

3.3 PixMo-SceneGraph (PixMo-SG)

After filtering and postprocessing, we obtain 704K image-scene graph pairs: the PixMo-SceneGraph pretraining corpus. The resulting scene graphs contain an average of ∼1,200 tokens, or 2,200 tokens including augmentation, in comparison to PixMo-Cap captions of 200 tokens. Token-length distributions, including 1000 vision tokens per image, are provided in the Appendix.

4 Experiments

4.1 Model Training

Model. We follow the Molmo2 architecture (Clark et al., 2026), pairing a Qwen3 LLM (Yang et al., 2025) with a SigLIP2 So400m/14 vision encoder (Tschannen et al., 2025). Each image is encoded at 378x378 resolution as a resized global view along with up to 8 overlapping crops tiled over the image, which preserves fine-grained detail. For every image patch, we concatenate the encoder's third-to-last and ninth-to-last layer features. A connector turns these into visual tokens for the LLM. It first applies attention pooling over each 2x2 window of patches, using the window's mean feature as the query. It then projects each pooled feature into the language model's embedding space with a SwiGLU MLP. Visual tokens are inserted into the text sequence between image start and end tokens, with a separator token marking the end of each row. Training details in Appendix.

Pretraining. We pretrain on 704K images paired with scene graph (SG), captions (Cap), or both (SG+Cap), without and with augmentation (SG_{Aug}), for 4 epochs. Scene graphs are longer than captions, so to prevent them from dominating the token-level loss, we reweight each response's tokens by $2/\sqrt{n}$, where n is its number of target tokens. When an image has k annotations, each one is further downweighted by $1/\sqrt{k}$. For an image paired with both a caption and a scene graph, they are treated as independent data points rather than concatenated and conditioned on each other.

Finetuning. Each pretrained model is then finetuned identically on a mixture of academic vision-language datasets: TextVQA (Singh et al., 2019), ScienceQA (Lu et al., 2022), OKVQA (Marino et al., 2019), VQAv2 (Goyal et al., 2017), ChartQA (Masry et al., 2022), AI2-Diagram (Kembhavi et al., 2016), AOKVQA (multiple choice and direct answer) (Schwenk et al., 2022), and TallyQA (Acharya et al., 2019). Together, these cover general visual understanding, OCR, knowledge-based reasoning, counting, and chart and diagram understanding. We train for 14K steps at a batch size of 64, about 3 epochs per dataset on average. Following Molmo, each dataset is sampled in proportion to the square root of its size. TallyQA is capped at an effective size of 100K to prevent oversampling. Finetuning uses standard, unweighted token-level loss.

Table 1: **In-domain benchmark results across model scales.** Molmo-SceneGraph outperforms Qwen3-VL at all scales, LLaVA-OneVision-1.5 and -2.0 8B, Molmo2 8B, and Qwen3.5 9B.

	Text VQA	Science QA	OK VQA	VQAv2	Chart VQA	AI2D	A-OK VQA-MC	A-OK VQA-DA	TallyQA	Average
2B										
SG	87.7	95.8	73.0	90.9	75.0	91.5	85.9	69.0	80.5	83.2
SG+Cap	88.5	96.3	73.6	91.4	76.0	91.9	86.0	69.6	81.1	**83.8**
SG_{Aug}	86.9	95.7	74.2	90.7	75.0	91.3	85.2	69.1	80.7	83.2
SG_{Aug}+Cap	88.0	95.8	73.8	91.3	76.1	92.0	86.1	69.4	81.2	<u>83.7</u>
Cap	88.5	96.2	73.0	91.0	75.8	91.2	85.2	69.2	81.0	83.5
Qwen3-VL	88.3	87.3	63.6	87.7	80.0	88.0	83.2	58.6	77.8	79.4
4B										
SG	88.9	98.1	76.8	92.2	80.1	94.9	88.6	72.8	82.2	86.1
SG+Cap	89.9	98.0	76.9	92.6	81.0	95.3	87.7	72.5	82.6	**86.3**
SG_{Aug}	88.7	98.1	77.4	92.3	80.2	95.0	86.9	71.9	82.1	85.9
SG_{Aug}+Cap	90.0	98.0	77.5	92.7	80.8	94.7	87.6	72.4	82.8	**86.3**
Cap	89.8	98.3	76.5	92.6	80.6	94.5	88.1	72.3	82.7	<u>86.2</u>
Qwen3-VL	90.3	92.1	70.4	90.4	84.1	92.8	86.0	63.6	81.2	83.4
8B										
SG+Cap	90.3	98.3	77.8	92.9	82.0	95.3	87.8	73.6	83.2	<u>86.8</u>
SG_{Aug}+Cap	89.7	98.2	78.6	93.0	81.2	95.9	88.6	73.5	82.9	<u>86.8</u>
Cap	90.0	98.5	77.7	92.8	82.0	95.7	88.6	73.6	83.0	**86.9**
Qwen3-VL	90.9	94.5	74.0	91.0	85.4	94.1	88.3	65.7	81.8	85.1
Other strong VLMs										
LLAVA-OV-1.5 8B	87.7	98.7	76.0	90.5	86.2	94.2	88.9	68.1	80.2	85.6
LLAVA-OV-2 8B	86.3	98.2	73.7	89.8	86.0	94.3	88.2	63.9	79.2	84.4
Molmo2 8B	84.9	98.0	71.1	87.7	72.5	96.0	90.0	65.6	73.5	82.1
Qwen3.5 9B	91.8	95.5	73.1	91.5	86.3	94.5	90.2	67.5	82.7	85.9

4.2 Model evaluation

We evaluate Molmo-SceneGraph in two settings. The first is in-domain, on the academic datasets the models were finetuned on. The second is out-of-domain, on benchmarks unseen during training. The out-of-domain set includes general-purpose benchmarks: RealWorldQA (xAI, 2024); MM-Bench (Liu et al., 2024); and SEED-Bench (Li et al., 2024), as well as benchmarks that target grounding and reasoning: VSR (Liu et al., 2023a) tests spatial reasoning; VisualCOMET (Park et al., 2020) tests temporal reasoning; Reporting Bias (Kamath et al., 2026) has separate splits for spatial reasoning, negation, temporal reasoning, and counting; and POPE (Li et al., 2023) measures object hallucination with yes/no questions such as "Is there a banana in the image?". We report accuracy on all benchmarks except POPE, for which we report F1. We evaluate on test splits of benchmarks where available, and validation splits otherwise. Results are shown in Tables 1 and 2.

We modify two benchmarks before evaluation: for VisualCOMET, we convert the before/after event annotations into multiple choice questions; for Reporting Bias Temporal, we expand the benchmark from 200 to 568 examples to increase evaluation variance. Details are provided in the Appendix.

5 Findings

5.1 Structuring pretraining data improves generalization

Comparing the Molmo-SceneGraph models pretrained on unaugmented scene graphs alone with the models pretrained on captions alone, we find that structuring the pretraining data largely preserves in-domain performance while improving out-of-domain performance. In particular, scene-graph pretraining improves performance on previously unseen grounding and reasoning tasks, despite using no additional information beyond that contained in the original captions. The structuring of the same pretraining data may enable clearer compartmentalization of the model's skills, allowing it to better generalize to new skills; as discussed in the context of humans in Section 3.1.

Table 2: **Out-of-domain benchmark results across model scales.** Molmo-SceneGraph outperforms Qwen3-VL at 2B and 4B, particularly on grounding and reasoning benchmarks, and LLaVa-OneVision-1.5 8B and -2.0 8B and Molmo2 8B, but falls short of Qwen3-VL 8B and Qwen3.5 9B.

	RealWorld QA	MM Bench	SEED-Bench	VSR	Visual Comet	RB-Spatial	RB-Negation	RB-Temporal+	RB-Counting	POPE	Avg.
2B											
SG	67.6	82.7	76.7	78.8	45.4	90.0	29.9	80.8	88.2	88.4	72.9
SG+Cap	67.8	83.4	76.6	78.4	44.7	94.5	35.2	78.6	90.0	88.7	73.8
SG_{Aug}	67.2	83.6	76.6	77.5	56.2	95.6	31.9	85.2	86.8	88.4	<u>74.9</u>
SG_{Aug}+Cap	67.9	83.6	76.9	79.4	52.5	98.1	32.4	83.5	89.2	88.3	**75.2**
Cap	66.7	83.6	76.1	74.9	44.9	91.5	34.7	76.0	89.8	88.6	72.7
Qwen3-VL	66.4	83.7	76.1	75.0	48.4	97.1	30.0	78.7	87.4	88.9	73.2
4B											
SG	71.2	86.5	77.9	80.7	64.6	99.0	54.7	86.4	88.6	88.8	79.9
SG+Cap	71.8	87.6	78.4	82.4	58.7	99.2	53.2	90.3	89.8	89.0	80.0
SG_{Aug}	69.8	86.5	77.8	81.3	67.0	99.3	51.4	90.5	89.0	89.0	<u>80.2</u>
SG_{Aug}+Cap	72.5	87.3	78.2	82.1	66.4	99.4	52.8	90.2	89.0	88.7	**80.7**
Cap	72.5	86.6	78.5	79.3	59.4	99.4	50.1	87.3	91.0	88.7	79.3
Qwen3-VL	72.7	89.3	79.0	83.7	66.6	99.3	48.4	86.8	84.2	89.1	79.9
8B											
SG+Cap	72.2	87.4	78.7	82.6	66.6	99.0	52.1	87.9	90.3	88.4	80.5
SG_{Aug}+Cap	71.2	87.9	78.7	81.7	69.8	99.3	53.0	92.4	90.5	88.4	<u>81.3</u>
Cap	72.5	87.0	78.5	80.4	67.0	98.8	51.9	90.1	89.9	88.8	80.5
Qwen3-VL	70.1	90.0	79.1	86.0	72.8	99.5	51.9	91.5	90.1	88.1	**81.9**
Other strong VLMs											
LLAVA-OV-1.5 8B	68.6	88.8	77.0	78.2	69.2	93.7	51.2	86.8	82.2	88.2	78.4
LLAVA-OV-2 8B	69.9	88.7	78.4	79.5	71.0	98.3	48.0	89.4	86.2	87.1	79.7
Molmo2 8B	77.6	87.8	75.1	80.0	60.8	99.3	41.5	85.0	89.2	88.3	78.5
Qwen3.5 9B	77.0	91.6	80.0	85.8	73.4	99.5	60.8	92.3	87.6	89.1	**83.7**

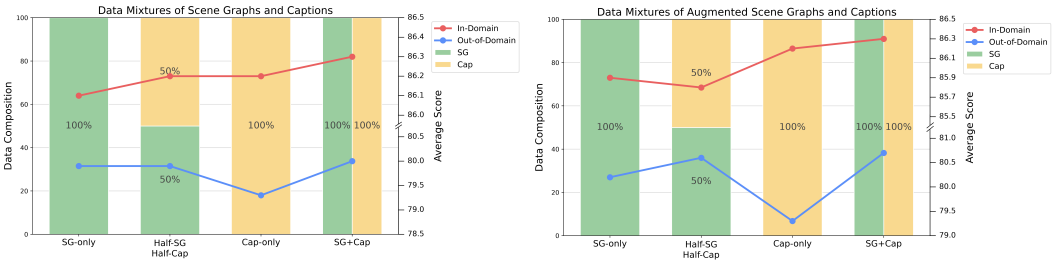
5.2 Combining structured and unstructured pretraining data improves performance

We next investigate the combination of structured and unstructured pretraining data. We find that combining the two forms of pretraining data—pairing each image with both its caption and its unaugmented scene graph—outperforms both caption-only and scene graph-only pretraining, both in-domain and out-of-domain. At 2B parameters, the combined model improves over caption-only pretraining by 0.4 points in-domain and 1.1 points out-of-domain; at 4B parameters, the corresponding improvements are 0.1 and 0.7 points. Potentially, having the captions at hand may allow the model to better parse the scene graph data, as the LLM is pretrained on natural language text, although we leave investigation of this mechanism to future work.

5.3 Augmenting the scene graph improves generalization

The structured representation also provides a convenient mechanism for incorporating information that tends to be absent in the original captions. As described in Section 3.2, we augment the scene graphs with temporal statements, negations, external knowledge, affordances, and agent information using dedicated fields in the scene-graph representation. We find that this additional information has little effect on in-domain performance, but substantially improves out-of-domain performance, particularly on grounding and reasoning tasks.

At 2B parameters, augmenting the scene graph improves the scene-graph-only model by 2.0 points overall out-of-domain and by 2.9 points on grounding and reasoning tasks. When captions and scene graphs are combined, the corresponding improvements are 1.4 and 1.9 points. At 4B parameters, the scene-graph only model improves on out-of-domain tasks after augmentation by 0.3 points overall, and 0.6 points on the grounding and reasoning tasks. The combined model improves by 0.6 and 0.9 points, respectively. At 8B parameters, where we train only the combined model, augmenting the scene graphs improves out-of-domain performance by 0.8 points overall and 1.2 points on grounding and reasoning tasks. Having open-sourced the method to generate synthetic captions, we leave scaling up the data to future work to potentially unlock greater gains at higher parameter scales.



(a) Without augmentation (b) With augmentation

Figure 3: The impact of different pretraining mixtures on Molmo-SceneGraph 4B. Mixing half of each type of data falls behind combining all of both, but could be explored for data efficiency.

These results are consistent with the reporting-bias hypothesis (Gordon & Van Durme, 2013; Misra et al., 2016; Kamath et al., 2026): information that is omitted from captions can be important for downstream grounding and reasoning, even when it is implicit or "obvious" to human annotators. Explicitly providing this information during pretraining improves performance on tasks that require it, demonstrating that the structured representation can serve not only to reorganize existing supervision but also as a flexible interface for adding targeted information.

5.4 Our models outperform Qwen3VL and other strong VLMs

We compare Molmo-SceneGraph with Qwen3-VL as both use a Qwen3 language-model backbone and SigLIP2 vision encoder, and Qwen3-VL has models at 2B, 4B and 8B. Across model scales, our models achieve substantially stronger performance than Qwen3-VL on the in-domain benchmarks, despite being trained on only 704K images. Qwen3-VL is trained through a four-stage pipeline comprising more than 2.17T image and text tokens Bai et al. (2025). While exact training data composition is not public, the authors report using data spanning knowledge, OCR, document parsing, grounding and counting, spatial understanding, science, and multimodal reasoning. Thus, its training data encompasses capabilities evaluated by both our in-domain and out-of-domain benchmarks. In contrast, Molmo-SceneGraph variants are pretrained on 3.4–9.6B tokens and fine-tuned on an additional 0.75B tokens, for a total of 4.2–10.4B tokens. Our models therefore achieve stronger performance with orders of magnitude fewer training tokens and a much smaller image corpus.

Our strongest Molmo-SceneGraph model (SG_{Aug}+Cap) outperforms Qwen3-VL by 4.4, 2.9, and 1.8 points in-domain at the 2B, 4B, and 8B scales, respectively. On the out-of-domain benchmarks, our models outperform Qwen3-VL at the 2B and 4B scales by 2.0 and 0.8 points respectively, with particularly strong gains on grounding and reasoning tasks, on which they improve over Qwen3-VL by 2.5 and 1.5 points respectively. At 8B, however, Qwen3-VL performs better on the out-of-domain average by 0.6 points, highlighting that higher-parameter models may saturate our 704K image-text pretraining corpus, and suggesting future work with a larger pretraining corpus.

Molmo-SceneGraph outperforms LLaVA-OneVision-1.5 8B, LLaVA-OneVision-2.0, and Molmo2 8B in-domain and out-of-domain, and Qwen3.5 9B in-domain. These results demonstrate that high-quality, structured pretraining data can produce strong VLMs from a relatively small image corpus. This finding is consistent with observations from Molmo Deitke et al. (2024) that carefully curated data can support strong multimodal performance without relying solely on large-scale web data.

That Molmo-SceneGraph outperforms Molmo2 8B, despite having the same architecture, further highlights the value of structured pretraining data, and PixMo-SceneGraph as pretraining corpus.

6 Analysis

6.1 Mixtures of scene graph and caption data

To further explore different combinations of structured and unstructured pretraining data, we train variants of Molmo-SceneGraph on a different type of mixture of caption and scene graph data, where 50% of the images are paired with captions alone, and the other 50% with scene graphs alone. We

consider this setting both with and without augmentation of the scene graphs. Results averaged across the in-domain and out-of-domain benchmarks are shown in Figure 3.

Without augmentation, the model trained on an equal mixture of captions and scene graphs achieves the best of both worlds in-domain and out-of-domain, matching whichever of the full-caption and full-scene graph performances is higher. Further, its results are only slightly below that of the model trained completely on both captions and scene graph data.

With additional information, the equal mixture exhibits a different pattern. Its in-domain performance is slightly below that of both the full-caption and full-scene-graph models, but its out-of-domain performance actually exceeds both. Moreover, its out-of-domain performance nearly matches that of the model trained on the full combination of caption and scene graph data.

Together, these results suggest that mixtures of caption and scene graph data may offer additional opportunities for balancing in-domain and out-of-domain performance while reducing the number of training tokens, motivating further exploration of mixture ratios and data composition. Following our earlier hypothesis that the presence of captions may facilitate initial parsing of the scene graph by the text-pretrained LLM, curriculum training approaches may be worth exploring as well.

6.2 Emergent uncertainty estimation in Molmo-SceneGraph

PixMo-SceneGraph corpus was generated by an LLM from captions without access to the image. Given the discussed ambiguity of captions, there are times that the LLM is unable to resolve certain aspects of the caption into a scene graph (e.g., unclear coreference resolution); hence, we prompt the model to include an uncertainty estimate in its scene graph generation ("if an attribute is uncertain or inferred, mark it as 'possible' or 'likely'"). This culminated in an interesting emergent behavior in the Molmo-SceneGraph model trained on this data. Molmo-SceneGraph has access to the image, but in cases where it is uncertain, it uses the same language of uncertainty, or providing instantiation for its inferences. For example, in Figure 1, the full scene graph (c.f. Appendix) states that the dolly "likely" has wheels—upon closer inspection of the image, the base of the dolly is cropped out, explaining the model's stated uncertainty despite knowing that dollies often have wheels (generated in external knowledge). Molmo-SG's emergent uncertainty estimation calls for further study.

7 Discussion and Future Work

We show that structuring VLM pretraining data can reduce the ambiguity inherent to captions, as well as serve as an interface to augment the corpus with data critical to grounding and reasoning that is typically left out of captions. We curate PixMo-SceneGraph, a dataset with 704K image-scene-graph pairs, and train Molmo-SceneGraph models, which achieve strong performance relative to Qwen3-VL and other popular models across model scales despite being trained on only a small fraction of the tokens. Our results suggest that the structure and information content of VLM pretraining data can be as important as its scale, and motivate further investigation into alternatives to the standard image-caption pretraining paradigm.

Our work raises several questions for future investigation. First, while we show that structured representations can improve VLM pretraining, it remains unclear which types of structure are most beneficial for different downstream tasks. For example, incorporating synthetic images such as screenshots into the pretraining data, together with structured descriptions of their contents, could potentially improve performance on web-agent tasks after fine-tuning on datasets such as MolmoWeb (Gupta et al., 2026), as the structured pretraining would align with the structured nature of web data. Incorporating more detailed affordance information could improve performance on embodied tasks such as ALFRED (Shridhar et al., 2020). Second, we discuss the uncertainty estimation that emerges from our model's generated scene graphs as a function of its pretraining. Making this uncertainty explicit and calibrated would be an interesting avenue for future work. Third, our scene graphs do not incorporate explicit grounding supervision, such as pointing annotations from PixMo-Points, bounding box data from Visual Genome, segmentation information from SAM3 (Carion et al., 2026), depth information, 2D/3D geometry, and tracking. Such information could be readily incorporated into the scene-graph representation by leveraging its natural flexibility to augment different types of supervision, and has strong potential to further improve VLM grounding performance.

AI USE STATEMENT

In this work, we used AI to (1) generate the scene graph training data (GPT), which we meticulously postprocessed and sampled to verify quality (samples provided in Appendix); (2) add evaluations to our codebase (Claude), which the first author verified manually; (3) polish sections of our paper after we wrote it (GPT), which the authors then took multiple passes through; (4) find any related work we missed (GPT), which the authors then searched for and read. We have NOT used AI for: (1) actual ideation; (2) writing the paper; (3) model training code. We have reviewed all AI-assisted work. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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A APPENDIX

A.1 PROMPT TO GPT TO CREATE SCENE GRAPHS

```
[SYSTEM_PROMPT:]
You are an AI specialized in structured information extraction. Your task is to
transform natural language image descriptions into structured JSON representations
while preserving all explicitly mentioned details.

[PROMPT:]
You will be given a piece of text describing an image. Your task is to convert all
information about the scene into a structured JSON format. Follow these guidelines
strictly:

### 1. Core Structure
- **objects**: Identify and list all entities in the description, such as people,
  animals, objects, and environmental elements. If there are multiple of the same
  entity, list them separately.
- **object IDs**: Assign a unique object ID to each object, e.g., 'dog1'.
- **attributes**: Assign clear and objective attributes to each object, such as
  color, material, size, shape, texture, function, and condition. If an attribute
  is uncertain or inferred, mark it as 'possible' or 'likely'. If the object
  contains writing, add its contents as an attribute called 'OCR'.
- **agent**: If the object is an agent, include a dictionary of agentic attributes
  like identity, emotional states, goals, actions, gaze direction, and social role.
- **external knowledge**: Add at least 10 facts you have about each object, e.g.,
  if a named entity is mentioned, add information about it.
- **negations**: Add a top-level array of 10 negation statements that are
  definitely NOT true about the image (e.g., 'There are no cars in the image').
  Only mention something if you are completely sure it is not true.
- **temporal statements**: Add a top-level array of 10 statements that definitely
  happened before this image, or is likely to happen after this image.
- **counts**: Add a top-level array of counts of various types of objects, where
  applicable.
- **relationships**: Place all relationships in a top-level array named
  'relationships'. Each relationship should be an object containing 'subject',
  'relationship', and 'object' fields, referencing relevant objects from 'objects'
  by their 'object ID'. Include 'external knowledge' that might apply to the
  specific relationship. Avoid embedding relationships inside the objects
  themselves.
- **affordances**: Add a top-level array of detailing what specific objects,
  attributes, or relations afford the viewer of the scene.

- **reasoning** (optional): If the caption describes an image that is person-centric
  (key agent or multiple) agents, add a top-level array with information that can
  be added using deduction, abduction, and transformation with the completed scene
  graph. Deduction: based on existing elements, what new attributes and relations
  must be true. Abduction: based on the existing elements, what new attributes and
  relations can be inferred since they are probable. Transformation: what would
  happen if an element in the scene was changed, e.g., viewer position in space,
  point in time? Indicate what object(s) and/or relationship(s) underlie this
  inference in 'basis'. Relationships in basis should include both relevant
  objects in a list. Include 'support' to detail what attributes of the object or
  relationship support the inference.

### 2. Optional Structure
- **viewpoint / context**: Capture details about perspective, camera angle,
  lighting conditions, environment, weather, possible location, or time of day
  (e.g., 'overhead view', 'nighttime', 'first-person perspective').
- **notes**: Include any additional information that does not fit into objects,
  attributes, or relationships, such as inferred meanings, artistic interpretations,
  or metadata.

### 3. Key Requirements
- **Minimize linguistic ambiguity**: Use concise and objective terms to represent
  details. If any descriptions are subjective or uncertain, indicate this
  explicitly in the attributes.
- **Output only valid JSON**: Do not include explanations, commentary, or formatting
  outside the JSON structure.
- **Retain all details**: Do not omit any objects, attributes, or relationships
  from the original description.

### 4. Example Field Naming
- **objects**: An array where each element includes at least 'name', 'object ID',
  'attributes' and 'external knowledge', and may contain 'type' or 'category'.
- **relationships**: An array where each entry is a dictionary with 'subject',
  'relationship', and 'object'.
- **viewpoint / context**: Captures general scene setting, perspective, or
  environmental details.
- **notes**: Stores additional information that does not fit elsewhere.
```

```
### Task:
Convert the following image description into structured JSON format that adheres
to these guidelines.
Output **only** the JSON, printed in JSON format --- no additional explanation or
formatting outside of it.

Below is the original image description text:
```

A.2 POSTPROCESSING OF PIXMO-SCENEGRAPH

We apply several postprocessing steps to standardize the generated scene graphs and construct the final training data. First, we ensure that each output is valid JSON, including removing Markdown code fences (e.g., "'`json`) that are occasionally produced by the generation model. We then normalize inconsistent field names (e.g., "affordance" versus "affordances") and standardize object counts as dictionaries mapping object names to their corresponding counts. We also enforce a maximum of two external knowledge facts per object. Finally, we impose a consistent ordering on the elements of each scene graph.

A.3 FULL AUGMENTED SCENE GRAPH EXAMPLE FROM PIXMO-SCENEGRAPH (PRETRAINING DATA)

Here is the full scene graph from PixMo-SceneGraph, seen in Figure 2.


```
{
 "objects": [
  {
   "name": "two-lane city road",
   "object ID": "road1",
   "type": "infrastructure",
   "attributes": {
    "road_lanes": 2,
    "lane_direction": "same direction",
    "foreground_position": "foreground",
    "road_surface": "asphalt",
    "alignment": "continues beyond frame",
    "possible_state": "possible daytime street conditions (no debris described)",
    "color": "possible dark gray"
   },
   "external knowledge": [
    "Two-lane roads typically have traffic moving in the same direction when explicitly
        described.",
    "Asphalt road surfaces are commonly dark gray to black and durable."
   ]
  },
  {
   "name": "edge pole at sidewalk",
   "object ID": "pole1",
   "type": "street furniture",
   "attributes": {
    "position_relative_to_road": "right side of road",
    "location_edge": "edge of sidewalk",
    "shape": "vertical pole",
    "color": "possible gray/metal (not explicitly stated)",
    "visibility": "standing pole near road edge"
   },
   "external knowledge": [
    "Street poles are commonly used for lighting, signage, or utilities.",
    "Poles at sidewalks are typically stationary infrastructure."
   ]
  },
  {
   "name": "white vehicle (partly visible)",
   "object ID": "vehicle1",
   "type": "vehicle",
   "attributes": {
    "color": "white",
    "visibility": "partly visible",
    "position_in_scene": "rearmost lane",
    "driven_by": "unseen person",
    "motion": "possible moving (implied by scene of riding on road; not explicit)",
    "vehicle_type": "possible car (not explicitly stated)"
   },
   "external knowledge": [
```

```
      "Vehicles are typically driven by a driver inside the cabin (driver unseen).",
      "White vehicles are common in urban traffic."
    ]
  },
  {
    "name": "cityscape background",
    "object ID": "cityscape1",
    "type": "environment",
    "attributes": {
      "position": "background",
      "content": "city buildings",
      "style": "urban skyline"
    },
    "external knowledge": [
      "Cityscapes in photos provide context about location and density.",
      "Buildings in the background often appear smaller due to perspective."
    ]
  },
  {
    "name": "park area with small sculptures",
    "object ID": "park1",
    "type": "environment",
    "attributes": {
      "position": "background alongside cityscape",
      "features": "small sculptures",
      "adjacent_elements": "geometric patterned fence",
      "landscape": "park area"
    },
    "external knowledge": [
      "Parks in urban settings often contain sculptures or public art installations.",
      "Small sculptures can be used to decorate walkways or landscaped sections."
    ]
  },
  {
    "name": "geometric patterned fence",
    "object ID": "fence1",
    "type": "fence/structure",
    "attributes": {
      "position": "background near park area",
      "pattern": "geometric patterned",
      "color": "possible light gray/metal (not explicitly stated)",
      "material": "possible metal (not explicitly stated)"
    },
    "external knowledge": [
      "Geometric fences often use repeating shapes for decorative and functional purposes.",
      "Fences commonly serve to define boundaries and restrict access."
    ]
  },
  {
    "name": "tall trees",
    "object ID": "trees1",
    "type": "vegetation",
    "attributes": {
      "position": "background",
      "height": "tall",
      "count": "multiple (not specified)",
      "juxtaposition": "against towering apartment building"
    },
    "external knowledge": [
      "Trees in background provide natural contrast against buildings.",
      "Tall trees can be part of park landscaping."
    ]
  },
  {
    "name": "towering apartment building",
    "object ID": "building1",
    "type": "building",
    "attributes": {
      "position": "background",
      "stories": "at least 20 stories",
      "style": "high-rise apartment building",
      "height_description": "towering"
    },
    "external knowledge": [
      "High-rise apartment buildings commonly exceed 20 stories.",
      "Tall buildings indicate dense urban habitation."
    ]
  },
  {
    "name": "clear vibrant blue sky",
    "object ID": "sky1",
```

```
    "type": "sky/weather",
    "attributes": {
      "position": "top/background",
      "condition": "clear",
      "color": "vibrant blue"
    },
    "external knowledge": [
      "A clear blue sky indicates good visibility.",
      "Daytime lighting improves color saturation in outdoor scenes."
    ]
  },
  {
    "name": "vintage-style motorcycle",
    "object ID": "motorcycle1",
    "type": "motorcycle",
    "attributes": {
      "style": "vintage-style",
      "position": "foreground",
      "visible_parts": "rider and motorcycle components (specific parts not enumerated)",
      "surface_finish": "glimmers with shiny black components",
      "color": "shiny black",
      "materials": "metal (likely) with reflective finish",
      "shine": "high reflectivity"
    },
    "external knowledge": [
      "Vintage-style motorcycles commonly feature classic silhouettes and exposed metal
          components.",
      "Shiny black finishes are often achieved with polished metal or glossy paint (possible
          )."
    ]
  },
  {
    "name": "man riding motorcycle",
    "object ID": "rider1",
    "type": "person",
    "attributes": {
      "role": "motorcycle rider",
      "position": "foreground, on motorcycle",
      "pose": "riding (implied seated posture)",
      "outfit_theme": "medieval/fantasy-inspired armor and knight-like helmet",
      "jacket_or_shirt_color": "bright red",
      "jacket_or_shirt_visibility": "beneath armor",
      "gloves_color": "black",
      "pants_color": "gray underneath armor",
      "boots_color": "black",
      "facial_hair": "shadow or black mustache visible (uncertain which)",
      "sunglasses": "black sunglasses",
      "helmet_mohawk_effect": "giant feathered mohawk illusion via feathers atop metal post
          ",
      "armor_details": "numerous bolts and layers"
    },
    "agent": {
      "identity": "unidentified man",
      "emotional_states": [
        "possible excitement (implied by 'exhilarating scene' and dramatic costume)"
      ],
      "goals": [
        "travel along the road on the motorcycle"
      ],
      "actions": [
        "rides a vintage-style motorcycle"
      ],
      "gaze_direction": "possible forward (not explicitly stated)",
      "social_role": "standout performer/character in a public setting"
    },
    "external knowledge": [
      "Motorcycle riders typically sit astride the bike with hands on handlebars.",
      "Armor-style costumes commonly use layered plates and fasteners to create a knight
          aesthetic."
    ]
  },
  {
    "name": "fantasy armor (worn by rider)",
    "object ID": "armor1",
    "type": "clothing/armor",
    "attributes": {
      "worn_by": "rider1",
      "style": "fantasy-inspired elaborate armor",
      "structure": "numerous bolts and layers",
      "color": "metallic tones (implied by shiny metallic look)",
      "coverage": "covers torso and extends over body (extent not fully enumerated)",
```

```
      "condition": "appears intact"
    },
    "external knowledge": [
      "Fantasy armor props often include layered components for depth and realism.",
      "Bolts and rivets are frequently used as decorative and structural details."
    ]
  },
  {
    "name": "bright red jacket or shirt (under armor)",
    "object ID": "clothing_red1",
    "type": "clothing",
    "attributes": {
      "worn_by": "rider1",
      "color": "bright red",
      "position": "beneath armor",
      "visibility": "contrasting visibility under metallic armor"
    },
    "external knowledge": [
      "Bright red fabrics often stand out strongly against metallic and dark tones.",
      "Wearing contrasting colors under armor is common in costume design."
    ]
  },
  {
    "name": "black gloves (worn by rider)",
    "object ID": "gloves1",
    "type": "clothing",
    "attributes": {
      "worn_by": "rider1",
      "color": "black",
      "material": "possible leather or synthetic (not specified)",
      "role": "hand protection / costume element"
    },
    "external knowledge": [
      "Motorcycle gloves are commonly used for grip and protection.",
      "Black gloves match many motorcycle and costume aesthetics."
    ]
  },
  {
    "name": "gray pants under armor (worn by rider)",
    "object ID": "pants1",
    "type": "clothing",
    "attributes": {
      "worn_by": "rider1",
      "color": "gray",
      "position": "under the armor",
      "visible_state": "visible underneath armor (extent not specified)"
    },
    "external knowledge": [
      "Base pants under armor can provide comfort and support.",
      "Gray is a neutral tone commonly used for underlayers."
    ]
  },
  {
    "name": "black boots (worn by rider)",
    "object ID": "boots1",
    "type": "clothing",
    "attributes": {
      "worn_by": "rider1",
      "color": "black",
      "footwear_type": "boots",
      "position": "on motorcycle (implied)",
      "condition": "appears intact"
    },
    "external knowledge": [
      "Boots help protect feet and ankles while riding (general knowledge).",
      "Black boots are common in both motorcycle gear and fantasy costumes."
    ]
  },
  {
    "name": "handmade knight-like helmet (worn by rider)",
    "object ID": "helmet1",
    "type": "headgear",
    "attributes": {
      "worn_by": "rider1",
      "construction": "handmade, unique",
      "shape": "rounded top",
      "ear_coverage": "separate metal sections covering ears",
      "neck_guard": "neck guard extends down from back of helmet",
      "color": "metallic tones (implied by metal sections)",
      "decor": "metallic mohawk accent on top (distinct from feather mohawk post)",
      "feature": "old-timey knight's headgear reminiscent"
```

```
    },
    "external knowledge": [
      "Handmade helmets in cosplay often feature custom-fit metal or composite parts.",
      "Rounded-top helmet shapes are common in knight-inspired designs."
    ]
  },
  {
    "name": "metal post rising from behind helmet",
    "object ID": "post1",
    "type": "head accessory/prop",
    "attributes": {
      "worn_by": "rider1 (attached behind head/helmet)",
      "position": "rises from behind his head",
      "angle": "bent forward at nearly a 90-degree angle",
      "height_relative": "at least a foot and a half above helmet",
      "color": "metallic",
      "purpose": "supports feather arrangement"
    },
    "external knowledge": [
      "Crest-like posts are common in fantasy helmets to support ornamental elements.",
      "A forward-bent post can improve visual prominence from the camera angle."
    ]
  },
  {
    "name": "feather arrangement (giant feathered mohawk)",
    "object ID": "feathers1",
    "type": "head accessory/prop",
    "attributes": {
      "worn_by": "rider1 (attached to post1)",
      "color": "vibrant, beautiful feathers",
      "shape_effect": "giant feathered mohawk illusion",
      "height_above_helmet": "at least a foot and a half above his helmet",
      "condition": "appears intact and full"
    },
    "external knowledge": [
      "Feathered headpieces create dramatic silhouettes in photos.",
      "A mohawk-like arrangement can be achieved by stacking or attaching feathers along a
          crest line."
    ]
  },
  {
    "name": "black sunglasses (worn by rider)",
    "object ID": "sunglasses1",
    "type": "eyewear",
    "attributes": {
      "worn_by": "rider1",
      "color": "black",
      "style": "sunglasses",
      "function": "eye protection (likely) / costume styling",
      "visibility": "visible on face"
    },
    "external knowledge": [
      "Sunglasses reduce glare and help visibility outdoors (general).",
      "Black sunglasses commonly pair with dramatic costumes."
    ]
  }
],
"relationships": [
  {
    "subject": "rider1",
    "relationship": "rides",
    "object": "motorcycle1",
    "external_knowledge": [
      "A rider on a motorcycle is typically seated and controlling the vehicle."
    ]
  },
  {
    "subject": "motorcycle1",
    "relationship": "is in the foreground",
    "object": "road1",
    "external_knowledge": [
      "Foreground vehicle placement is consistent with street-level photography."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "armor1",
    "external_knowledge": [
      "Armor-like costumes are worn over base clothing."
    ]
```

```
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "helmet1",
    "external_knowledge": [
      "Helmets are worn on the head and can anchor decorative elements."
    ]
  },
  {
    "subject": "helmet1",
    "relationship": "is complemented by",
    "object": "post1",
    "external_knowledge": [
      "A crest post can be attached to or rise from the helmet area."
    ]
  },
  {
    "subject": "post1",
    "relationship": "supports",
    "object": "feathers1",
    "external_knowledge": [
      "Decorative posts often hold feather arrangements."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "sunglasses1",
    "external_knowledge": [
      "Sunglasses are facial accessories worn on the rider."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "gloves1",
    "external_knowledge": [
      "Gloves are worn on hands and typically help grip and safety."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "pants1",
    "external_knowledge": [
      "Pants can serve as underlayer beneath armor."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "wears",
    "object": "boots1",
    "external_knowledge": [
      "Boots are worn on the feet and support riding stance."
    ]
  },
  {
    "subject": "rider1",
    "relationship": "has under-armor clothing",
    "object": "clothing_red1",
    "external_knowledge": [
      "Base layers beneath armor can show through openings."
    ]
  },
  {
    "subject": "vehicle1",
    "relationship": "is in",
    "object": "road1",
    "external_knowledge": [
      "Vehicles travel on road lanes."
    ]
  },
  {
    "subject": "pole1",
    "relationship": "stands on the",
    "object": "road1",
    "external_knowledge": [
      "Street poles commonly stand near road edges rather than in the driving lanes."
    ]
  },
```

```
    {
      "subject": "park1",
      "relationship": "is alongside",
      "object": "cityscape1",
      "external_knowledge": [
        "Parks often adjoin urban districts and are placed near roads."
      ]
    },
    {
      "subject": "park1",
      "relationship": "contains",
      "object": "fence1",
      "external_knowledge": [
        "Fences can be part of park perimeter or internal sections."
      ]
    },
    {
      "subject": "park1",
      "relationship": "is accompanied by",
      "object": "trees1",
      "external_knowledge": [
        "Parks commonly have trees for landscaping."
      ]
    },
    {
      "subject": "cityscape1",
      "relationship": "includes",
      "object": "building1",
      "external_knowledge": [
        "A cityscape can be defined by its prominent buildings."
      ]
    },
    {
      "subject": "building1",
      "relationship": "is adjacent to",
      "object": "trees1",
      "external_knowledge": [
        "Landscaping often sits near building grounds."
      ]
    },
    {
      "subject": "sky1",
      "relationship": "is above",
      "object": "cityscape1",
      "external_knowledge": [
        "Sky typically occupies the upper region of outdoor images."
      ]
    }
  ],
  "counts": {
    "road_lanes": 2,
    "apartment_buildings_mentioned": 1,
    "feathered_headpieces": 1,
    "motorcycles": 1,
    "people_visible": 1,
    "vehicles_visible": 1,
    "poles_mentioned": 1
  },
  "temporal statements": [
    "The rider was already wearing the elaborate armor and helmet before entering the roadway
         (likely).",
    "The motorcycle and rider are currently on the road at the moment the photograph was
        taken (implied by framing).",
    "A city street environment with park and high-rise buildings existed before the photo was
         captured (likely).",
    "The sky was clear at the time the photo was captured (explicit).",
    "The white vehicle existed on the roadway before the moment depicted (likely).",
    "The feathers and metal post were attached to the helmet before the rider departed (
        likely).",
    "After the photo, the rider will likely continue traveling along the road (likely).",
    "After the photo, viewers may comment on the costume and helmet (implied by 'likely to
        garner admiration').",
    "After the photo, the vehicles may continue moving forward in their lanes (likely).",
    "After the photo, the scene's lighting conditions will likely remain generally consistent
         if the weather stays clear (possible)."
  ],
  "negations": [
    "There are no pedestrians directly interacting with the rider.",
    "There is no visible crash or collision between vehicles.",
    "There is no rainy or snowy weather in the scene.",
    "There are no clouds covering the sky.",
```

```
    "There is no visible advertisement text on the fence or buildings.",
    "There is no visible motorcycle brand logo or model name.",
    "There is no visible driver inside the white vehicle.",
    "There is no visible license plate text on the white vehicle.",
    "There is no visible smoke, fire, or exhaust cloud dominating the scene.",
    "There is no visible helicopter or aircraft above the scene."
  ],
  "affordances": [
    "The road provides a travel path for vehicles and riders.",
    "The motorcycle affords rapid movement along the lane for the rider.",
    "The rider's armor and helmet afford high visual distinctiveness, likely attracting
        attention.",
    "The black sunglasses afford glare reduction and obscure eye direction in photos.",
    "The feathered mohawk post/feathers afford dramatic height and flamboyant silhouette.",
    "The roadside pole affords structural reference and possible utility/signage support.",
    "The park sculptures/fence afford aesthetic background texture and a curated public-space
         feel.",
    "Tall trees afford natural contrast against urban architecture.",
    "The clear blue sky affords strong daylight contrast enhancing reflective surfaces."
  ],
  "viewpoint/context": {
    "camera_angle": "likely eye-level or slightly low (not explicitly stated)",
    "lighting": "daytime natural light",
    "motion_sense": "rider is prominent; vehicles appear in lanes",
    "perspective": "street-level, foreground-to-background depth",
    "setting": "city road adjacent to park and high-rise buildings",
    "time_of_day": "possible daytime (exact time not specified)",
    "weather": "clear"
  },
  "notes": [
    "The description emphasizes a 'one-of-a-kind' medieval-inspired costumed rider and
        suggests the photo is intended to attract admiration.",
    "Uncertainty: the mustache is described as either a shadow or a black mustache; exact
        facial hair identity is not fully determined."
  ]
}
```


A.4 Full augmented scene graph example from Molmo-SceneGraph (Model Output)

Here is the full scene graph from the Molmo-SG output, seen in Figure 1


```
{
   "objects": [
      {
         "name": "man",
         "object ID": "man1",
         "type": "person",
         "attributes": {
            "clothing": "blue shirt",
            "headwear": "none mentioned",
            "pose": "pushing a dolly",
            "gaze_direction": "possible toward dolly/forward (not explicitly stated)",
            "emotional_state": "possible neutral/focused (not explicitly stated)",
            "role": "worker transporting bananas"
         },
         "agent": {
            "identity": "unidentified man",
            "social_role": "worker",
            "goals": [
               "transport bananas from truck to another location"
            ],
            "actions": [
               "pushing a dolly filled with bananas"
            ],
            "gaze_direction": "possible toward dolly/forward",
            "emotional_states": [
               "possible neutral/focused"
            ]
         },
         "external knowledge": [
            "A person pushing a dolly typically exerts forward force to move the load.",
            "Workers transporting produce often coordinate with another person to manage the
                dolly."
         ]
      },
      {
         "name": "man",
```

```
        "object ID": "man2",
        "type": "person",
        "attributes": {
          "clothing": "gray shirt",
          "headwear": "black baseball cap",
          "pose": "standing beside dolly",
          "gaze_direction": "possible toward dolly/forward (not explicitly stated)",
          "emotional_state": "possible neutral/focused (not explicitly stated)",
          "role": "worker assisting with banana transport"
        },
        "agent": {
          "identity": "unidentified man",
          "social_role": "worker",
          "goals": [
            "assist in transporting bananas"
          ],
          "actions": [
            "standing beside dolly"
          ],
          "gaze_direction": "possible toward dolly/forward",
          "emotional_states": [
            "possible neutral/focused"
          ]
        },
        "external knowledge": [
          "A baseball cap can provide sun protection outdoors.",
          "Standing beside a dolly often indicates supervision, coordination, or readiness
              to assist."
        ]
      },
      {
        "name": "dolly",
        "object ID": "dolly1",
        "type": "transport equipment",
        "attributes": {
          "contents": "filled with green bananas",
          "color": "not explicitly stated",
          "material": "possible metal or plastic (not explicitly stated)",
          "shape": "possible wheeled platform (not explicitly stated)",
          "condition": "not explicitly stated",
          "function": "carry bananas from truck to another location"
        },
        "external knowledge": [
          "A dolly is commonly used to move heavy or bulky items with reduced manual
              effort.",
          "Dollies often have wheels to roll over uneven ground."
        ]
      },
      {
        "name": "bananas",
        "object ID": "bananas1",
        "type": "food/produce",
        "attributes": {
          "color": "green",
          "quantity": "many (not explicitly countable)",
          "state": "fresh (implied by 'freshly picked')",
          "placement": "on dolly1",
          "visibility": "clearly visible as a load"
        },
        "external knowledge": [
          "Green bananas are typically harvested before full ripening.",
          "Bananas are commonly transported in bulk to markets or processing facilities."
        ]
      },
      {
        "name": "truck",
        "object ID": "truck1",
        "type": "vehicle",
        "attributes": {
          "color": "blue",
          "position": "behind the men",
          "function": "source of bananas being transported",
          "visibility": "partially visible (implied by 'behind them')",
          "condition": "not explicitly stated"
        },
        "external knowledge": [
          "A truck is commonly used to transport produce from farms to markets.",
          "Blue trucks are common in many regions and can be used for visibility."
        ]
      },
      {
```

```
        "name": "building",
        "object ID": "building1",
        "type": "building",
        "attributes": {
           "color": "white",
           "position": "left side of background",
           "windows": "one window with a red awning",
           "awning_color": "red",
           "condition": "not explicitly stated"
        },
        "external knowledge": [
           "Buildings with awnings often provide shade and weather protection.",
           "A red awning can indicate a storefront, office, or utility building."
        ]
     },
     {
        "name": "building",
        "object ID": "building2",
        "type": "building",
        "attributes": {
           "color": "yellow",
           "position": "right side of background",
           "windows": "two blue windows",
           "condition": "not explicitly stated"
        },
        "external knowledge": [
           "Yellow buildings are common in many urban and rural areas.",
           "Blue windows can be decorative or functional glazing."
        ]
     },
     {
        "name": "building",
        "object ID": "building3",
        "type": "building",
        "attributes": {
           "color": "white",
           "position": "right side of background (additional white building)",
           "windows": "not explicitly stated",
           "condition": "not explicitly stated"
        },
        "external knowledge": [
           "Additional buildings in the background suggest a developed area.",
           "White facades can reflect sunlight and appear bright in daylight."
        ]
     }
  ],
  "relationships": [
     {
        "subject": "man1",
        "relationship": "is pushing",
        "object": "dolly1",
        "external knowledge": [
           "Pushing a dolly typically involves applying forward force to move the load."
        ]
     },
     {
        "subject": "dolly1",
        "relationship": "is filled with",
        "object": "bananas1",
        "external knowledge": [
           "Dollies used for produce often carry multiple bunches or a bulk load."
        ]
     },
     {
        "subject": "man2",
        "relationship": "is standing beside",
        "object": "dolly1",
        "external knowledge": [
           "Standing beside a dolly often indicates coordination or readiness to assist."
        ]
     },
     {
        "subject": "truck1",
        "relationship": "is behind",
        "object": "man1",
        "external knowledge": [
           "In transport scenes, the source vehicle is often positioned behind workers."
        ]
     },
     {
        "subject": "truck1",
```

```
            "relationship": "is behind",
            "object": "man2",
            "external knowledge": [
               "Workers typically move from a truck to a destination while the truck remains
                    behind."
            ]
         },
         {
            "subject": "truck1",
            "relationship": "is the source of",
            "object": "bananas1",
            "external knowledge": [
               "The description explicitly states the bananas are being transported from the
                    truck."
            ]
         },
         {
            "subject": "building1",
            "relationship": "is located to the left of",
            "object": "building2",
            "external knowledge": [
               "The description places the white building with a red awning on the left and the
                     yellow building with two blue windows on the right."
            ]
         },
         {
            "subject": "building2",
            "relationship": "is located to the right of",
            "object": "building1",
            "external knowledge": [
               "The description places the yellow building on the right side of the background
                    ."
            ]
         },
         {
            "subject": "building3",
            "relationship": "is located to the right of",
            "object": "building1",
            "external knowledge": [
               "The description indicates an additional white building on the right side of the
                     background."
            ]
         }
      ],
      "counts": {
         "person": 2,
         "dolly": 1,
         "truck": 1,
         "buildings": 3,
         "windows_on_building2": 2
      },
      "temporal statements": [
         "The men are currently transporting bananas from the truck.",
         "The bananas were freshly picked before being loaded onto the dolly.",
         "The truck is positioned behind the workers at the time of the image.",
         "The dolly is being pushed forward during the scene.",
         "The workers likely will continue moving the dolly toward a destination after this
              moment.",
         "The bananas will likely be unloaded at a destination after transport begins.",
         "The buildings in the background existed before the image was taken.",
         "The truck likely arrived earlier than the moment captured to load the bananas.",
         "The workers likely coordinated their positions before pushing the dolly.",
         "The scene likely occurs during an active work period rather than after the work is
              complete."
      ],
      "negations": [
         "There are no visible animals in the image.",
         "There are no visible bicycles in the image.",
         "There are no visible motorcycles in the image.",
         "There are no visible cars in the image.",
         "There are no visible pedestrians other than the two men described.",
         "There is no visible text signage on the buildings mentioned.",
         "There is no visible fire or smoke in the scene.",
         "There is no visible water body (river, lake, or ocean) in the scene.",
         "There is no visible snow or ice in the scene.",
         "There is no visible night-time lighting; the scene is not described as nighttime."
      ],
      "affordances": [
         "The dolly affords rolling transport of the banana load across the ground.",
         "The bananas afford being handled and moved as a bulk produce item.",
         "The truck affords supplying bananas as a source of the transport load.",
```

```
      "The men's clothing (blue shirt, gray shirt, black baseball cap) affords
          identification of workers and sun protection.",
      "The red awning affords shade/protection for the window area on the left building.",
      "The blue windows afford visibility/lighting for the right-side building.",
      "The background buildings afford contextual cues about an urban or developed area."
   ],
   "viewpoint/context": {
      "background_elements": "buildings with windows and awnings",
      "camera_angle": "possible eye-level",
      "environment": "urban or developed area",
      "lighting": "possible daylight",
      "scene_description": "Two men are transporting bananas from a blue truck using a dolly
          ; background includes multiple buildings with windows and an awning.",
      "time_of_day": "possible daytime"
   },
   "notes": [
      "Exact dolly color/material and the men's exact gaze direction are not explicitly
          stated.",
      "The number of bananas is not explicitly countable; only that the dolly is filled with
           many green bananas.",
      "The exact time of day and lighting conditions are not explicitly stated.",
      "Two men in work attire push a dolly filled with green bananas from a blue truck;
          background includes buildings with a red awning and two blue windows."
   ]
}
```


A.5 TOKEN LENGTH DISTRIBUTIONS

The token distribution of Captions is shown in Figure 4. The token distribution of Scene Graphs without additional information is shown in Figure 5. The token distribution of Augmented Scene Graphs (with additional information) is shown in Figure 6. Note that the token count includes a standard 1000 vision tokens in all examples.

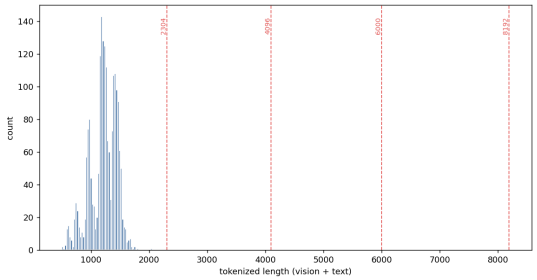


Figure 4: Token Distribution of Captions in PixMo-SG

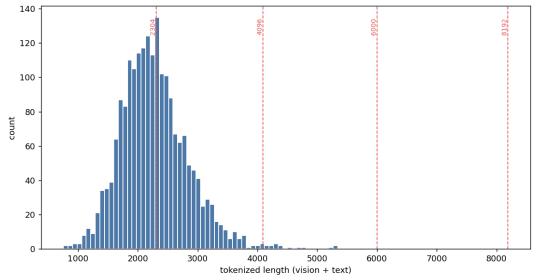


Figure 5: Token Distribution of Scene Graphs in PixMo-SG

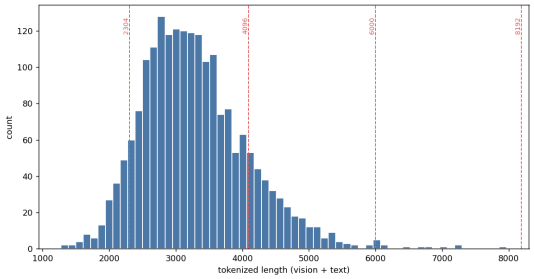


Figure 6: Token Distribution of Augmented Scene Graphs in PixMo-SG

A.6 Training details

All parameters, including the vision encoder, connector, and language model, are updated in both pretraining and finetuning. We train on 2 nodes of 8 NVIDIA H100 80GB GPUs, using bfloat16 mixed precision and FSDP. Both stages use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, no weight decay) with component-specific learning rates, gradient clipping at norm 1, a z-loss with weight 1e-4, and a linear warmup followed by cosine decay to 10% of the peak rate. In pretraining, we use a global batch size of 128 and maximum sequence lengths (including visual tokens) between 2100 for caption-only data and 4600 for data containing scene graphs with additional information. The newly initialized connector uses a learning rate of 2e-4 with a 200-step warmup. The language model uses 2e-5 and the vision encoder 6e-6, both with 2000-step warmups. In finetuning, we use a global batch size of 64, a maximum sequence length of 1536, learning rates of 5e-6 for the language model and 1e-6 for the connector and vision encoder, and a 200-step warmup for all components. We do not implement the bidirectional vision token attention or packing of Molmo2.

A.7 Evaluation details

We use standard metrics for all tasks. For AI2D, we report the transparent set. For ChartQA, we report relaxed correctness. For VQAv2, TextVQA, OKVQA and RealWorldQA we use an LLM judge to help with normalization (GPT-5.6-luna, which sees the answer and prediction only, not the question).

We modified two benchmarks before evaluating our models on them. The details are below:

VisualCOMET benchmark. We build a multiple-choice benchmark for temporal commonsense reasoning from the validation split of VisualCOMET (cite). Each VisualCOMET annotation pairs a movie still from VCR (cite) with a description of the current event, the intent of a person in the scene, and unordered lists of events that happened before and after the frame. People are referred to by numeric indices tied to bounding boxes. We keep the annotations that have at least three events in both lists (13,764 of 13,768) and randomly sample from them. For each sampled annotation, we give GPT-5.6-Terra (high reasoning effort) the image and the annotation. We ask it to write two questions: "What definitely happened before this image?" and "What most likely happened after this image?", based on the language used in the annotation collection of VisualCOMET. The model first ranks each list by how likely each event is to fall on its side of the depicted moment, judged from the image. For the before question, it takes the most likely "before" event as the answer and the three most likely "after" events as distractors, and it does the reverse for the after question. We instruct the model to make sure exactly one option is correct given the image, and to replace the numeric indices with natural descriptions (e.g., "the boy opened the door" instead of "2 open the door") so the questions are answerable from the raw frame, with no boxes drawn. Because the distractors come from the same scene's own event lists, they are topically plausible. Telling them apart requires reading the temporal state of the image, not rejecting unrelated events. We shuffle the answer options with a fixed per-annotation seed to remove position bias. We then manually review the generated questions and keep 500 unambiguous ones: four-way multiple-choice questions over 349 distinct images (319 before and 181 after questions), ensuring that no knowledge of the sequence of events in the corresponding movie is required to answer the question (e.g., scenes from the movie Titanic

that have "the ship will sink" as an "after" option although not directly implied by the image). We score models by accuracy.

Reporting Bias Temporal benchmark. The Temporal task of the Reporting Bias benchmark (**?**) contains only 200 examples. Each pairs a COCO image with two captions that differ only in "before" vs. "after" (e.g., "A person on the sidewalk before it has started to rain", "A person on the sidewalk after it has started to rain"), and the model must pick the caption that describes the image. To reduce evaluation variance, we extend the task with new examples built the same way. We sample 1000 COCO images that do not already appear in the benchmark. For each image, GPT-5.6-Terra (high reasoning effort) proposes up to two independent captions about two different events, one phrased with "before" and one with "after", together with a one-sentence justification for each. We accept a direction only if the image captures a single, unambiguous event or state change just before or just after it happens. The subject and the visual evidence must also be obvious at a glance and decidable from the image alone (e.g., rain visibly falling, an umbrella open vs. still closed, a plane in the air vs. on the runway). We instruct the model to mark a direction as unavailable rather than force a weak or ambiguous event, and to follow the caption style of the original benchmark. We discard images with no valid direction, which leaves 654 candidate images (225 with a "before" caption and 526 with an "after" caption). We then manually review the candidates and select 368 high-quality examples (184 "before" and 184 "after"), each on a distinct image. We pair each kept caption with a counterfactual that swaps "before" and "after" as the negative option for the multiple-choice question. We report accuracy on the union of the original and new examples, 568 in total.